Rohan Akurathi

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Education

University of California, Los Angeles (UCLA) — Los Angeles, CA
B.S. in Statistics and Data Science, GPA: 3.75 — *June 2026*

- **Relevant Coursework:** Machine Learning, Statistical Modeling, Predictive Modeling, Regression Analysis, Design of Experiments, Probability & Distributions, Algorithms & Data Structures

Technical Skills

Languages: Python (Pandas, NumPy, scikit-learn), SQL (PostgreSQL, MySQL/MariaDB, SQLite), R, JavaScript, C++
Machine Learning: PyTorch, XGBoost, LightGBM, CNNs, Reinforcement Learning (PPO/SAC), Logistic Regression, Random Forest, Clustering/Unsupervised, Time Series Forecasting, NLP, ONNX, DistilBERT, LLM Fine-tuning, Prompt Engineering, RAG, LangChain, Cross-Validation, ROC-AUC, Feature Engineering, Hyperparameter Tuning, SHAP
Data & Statistics: Hypothesis Testing, A/B Testing, Causal Inference, Experimental Design, Statistical Modeling, Regression Analysis
Visualization: Tableau, Power BI, Matplotlib, Seaborn
Tools: Git, Excel, Jupyter, Firebase, REST APIs, AWS, GCP, MLflow, FAISS, OpenCV, GitHub Copilot, Cursor, OpenAI/Anthropic APIs

Experience

Machine Learning Intern — May 2026 – Present
Occuspace — *Marina del Rey, CA*

- Train and retrain occupancy ML models in Python to launch new zones, tuning model parameters per site
- Diagnose faulty sensors by analyzing log and downtime patterns, surfacing root-cause guidance for operations
- Investigate data quality issues across sensor pipelines and deliver mitigation recommendations from large-scale logs
- Automate repetitive training and troubleshooting tasks in Python, improving technical operations efficiency

Undergraduate Researcher — Jan 2025 – Present
UCLA Center for Advanced Surgical & Interventional Technology — *Los Angeles, CA*

- Built scikit-learn pipelines (logistic regression, random forest, XGBoost) on clinical data, 0.89 ROC-AUC (+12%)
- Engineered 20+ features from health data via Pandas EDA/ETL, cutting model error 15% via feature selection
- Evaluated 3 architectures with 5-fold cross-validation; risk scores informed planning for 2 surgical studies
- Translated model outputs into risk scores and Matplotlib dashboards for clinical stakeholders across 2 studies

Project Lead — May 2025 – Sep 2025
UCLA Creative Labs — *Los Angeles, CA*

- Conceived and led an AI product across a 10-person design and engineering team, from concept to launch
- Optimized an LLM pipeline in PyTorch via batching and ONNX export, cutting inference latency 40%

ML & Data Engineering Developer — Jan 2025 – May 2025
UCLA Creative Labs — *Los Angeles, CA*

- Built real-time analytics for 2,000+ daily active users on Firebase, owning the schema behind 3 launches
- Built LLM recommendation and content-tagging models, raising personalization CTR 18% via A/B testing
- Reduced average API response latency 25% via request batching and caching in backend inference

Projects

Quantitative Trading Strategies with FinRL-X — *Mar 2026 – May 2026*
Python, Reinforcement Learning, Random Forest, Backtesting

- Built and backtested quant trading strategies on AI4Finance's FinRL-X, a modular weight-centric framework
- Implemented an ML stock-selection strategy with Random Forest scoring and PPO/SAC DRL allocation
- Backtested on the bt engine with transaction costs and no-lookahead splits, benchmarking Sharpe vs QQQ/SPY

Yelp Personalized Recommendation & Semantic Search System — *Jan 2026 – Mar 2026*
Python, LightGBM, FAISS, Sentence Transformers

- Built a two-stage recommender on 7M Yelp reviews with ALS and LightGBM ranking, +23% NDCG@10
- Built semantic search fine-tuning a sentence transformer over 7M+ FAISS embeddings, plus LLM taste profiles
- Engineered 30+ user/business features from review and check-in data, using SHAP to surface top ranking signals

Skin Cancer Classification via Hybrid Deep Learning — *Sep 2025 – Dec 2025*
Python, PyTorch, XGBoost, SHAP, OpenCV

- Built a hybrid classifier on 10,000+ ISIC/HAM10000 dermoscopic images, 0.91 ROC-AUC on malignant lesions
- Extracted visual features from a fine-tuned CNN backbone, feeding embeddings into a gradient-boosted classifier
- Applied SHAP pixel-mapping to validate feature importance and surface diagnostic hotspots for interpretability

Additional Information

Honors: Dean's List, Honors Certification, MC L.E.A.D.S. Certificate and Medallion
Interests: Gym, Basketball, Hiking, Cooking